A Distribution of Possibilities

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

A Distribution of Possibilities

> Prepare A Distribution of Possibilities: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Check whether a next-token table is a valid **probability distribution**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Predictive and Generative AI

Without opening the old lesson, explain one key idea from Predictive and Generative AI. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish discriminative **prediction** from generation by inspecting the requested **output**

Reference example, not the only acceptable wording: Teacher explanation: Message labels are discriminative **prediction**; continuing a sequence is generation; record lookup is retrieval rather than learned generation by itself.

One application can route a request, retrieve **evidence** and generate a response.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A toy next-token **distribution** is sun=.5, rain=.3, snow=.2. Over ten **sample**s, must sun occur exactly five times? Does the highest-probability token establish a true statement about weather?

Concept to explain

Week 2: A **Distribution** of Possibilities

Term	Meaning
distribution	A description of how values or probabilities are spread across possible outcomes.
likelihood	In this introductory context, how plausible an outcome is under a model; in statistics, likelihood has a more specific parameter-based meaning.
sample	A selected group of observations, or an outcome drawn from a distribution; the context identifies which meaning applies.
continuation	Text that follows an existing sequence.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Use two 45-minute sessions. All model **output**s and probability tables are constructed classroom examples. No hosted AI service, external API or student account is needed.

Distinguish a model objective from factual support and system permissions. Collect independent work before discussing its solutions.

Readiness answer

Different vocabularies and rules segment the same text differently; end markers may change the reported count.

Printed lesson and workbook; paper and pencil

Calculator; optional local Python 3 interpreter

Prepare the independent task and fresh retry without displaying solutions

Apply the connected idea: Embeddings and position

Use two 45-minute sessions. All model **output**s and probability tables are constructed classroom examples. No hosted AI service, external API or student account is needed.

Distinguish a model objective from factual support and system permissions. Collect independent work before discussing its solutions.

Readiness answer

No. IDs are indices. Similarity must refer to the **embedding** vectors and a defined comparison, not numerical ID distance.

Printed lesson and workbook; paper and pencil

Calculator; optional local Python 3 interpreter

Prepare the independent task and fresh retry without displaying solutions

End goal for the pictorial case

By the end, I can multiply conditional probabilities to compare paths instead of only first tokens.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model A Distribution of Possibilities using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Possible next words have weights: clear 5, useful 3, magical 1.

2. Nine tokens in the spinner represent those relative weights.

3. Repeated sampling can produce different sequences.

4. One **sample** does not reveal the whole **distribution** or guarantee truth.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

The teaching case

A toy next-token **distribution** is rain .5, sun .3, snow .2.

The teaching case

Check total and interpret the largest value.

Exact explanation to model

Why the result follows

Total is 1. Rain is most likely, not guaranteed under sampling. Token probability is not whole-sentence truth.

Explain the added worked example

Explain the added worked example · 1

A constructed model uses prefix “Status:” and vocabulary alternatives ready, busy and unknown. Its next-token table is ready 0.5, busy 0.3, unknown 0.2. All entries are nonnegative and sum to 1.

A probability belongs to a token under this prefix and model, not to that token in every situation. An omitted vocabulary alternative would make the interpretation incomplete unless its probability were explicitly zero.

Explain the added worked example · 2

A table containing 0.6,0.3,0.3 sums to 1.2 and is not already a **probability distribution**. If the values are nonnegative weights and the rule says normalise, divide each by 1.2. Do not silently treat weights as probabilities.

Negative weights cannot be repaired by this simple normalisation rule.

Explain the added worked example · 3

The probability of ready followed by end is 0.5*0.8 = 0.4. Ready followed by today is 0.5*0.2 = 0.1. Busy followed by end is 0.3*0.5 = 0.15, and busy followed by today is 0.15.

Unknown followed by end is 0.2 and unknown followed by today is 0. Together these six two-step paths sum to 1. These products are chain-rule calculations; they do not assume the second token is independent of the first.

Key terms

Term	Meaning
Probability distribution	Nonnegative probabilities over stated alternatives that sum to one
Conditional probability	A probability evaluated given specified information
Sequence probability	The product of successive conditional probabilities under an autoregressive model
Probability mass	The probability assigned to an alternative or a set of alternatives

Apply the connected idea: Embeddings and position

Apply the connected idea: Embeddings and position · 4

Use a two-token vocabulary: ID 1 means red with **embedding** [1,0]; ID 2 means blue with **embedding** [0,1]. This lookup table is constructed, not trained.

The **input** red blue has **embedding** rows [1,0] and [0,1], so its shape is two sequence positions by two components. Reversing the text reverses the rows. ID 2 is not “twice as blue” as ID 1 is red.

Apply the connected idea: Embeddings and position · 5

A model needs a way to use order. A collection of vectors without any order-sensitive procedure may lose that distinction. We introduce one explicit toy scheme: add **position vector** [0,0] at position 0 and [2,0] at position 1.

These invented values teach arithmetic; they are not a recommended production encoding.

Apply the connected idea: Embeddings and position · 6

The same red token now has [1,0] at position 0 and [3,0] at position 1. Both token identity and position affect the represented row.

Addition requires equal widths; adding a three-component **position vector** to a two-component **embedding** violates the **input** contract. Record the position numbering and **embedding**-table version.

Guided: A B gives [2,1],[2,2]; B A gives [1,3],[3,0]. Both sums [4,3]. The summation discarded the row-order distinction. Exit: ID gaps are not vector similarity; repeating positions violates the stated table range.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Model the reasoning aloud · 7

Greedy decoding chooses the largest next-token probability at each step, with an explicit tie rule if needed. It does not always maximise the probability of a complete sequence. Counterexample: first-step A=0.6 and B=0.4.

Given A, end=0.51 and x=0.49; given B, end=1. Greedy produces A,end with probability 0.306. The path B,end has probability 0.4, larger than 0.306.

This comparison concerns exactly two-step paths; longer completion probabilities require the later conditionals too.

A high path probability measures the model's preference over these specified paths. It is not **evidence** that a status claim matches a live **observation**. Format checks and source checks remain separate.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: A Distribution of Possibilities

A fictional two-stage token tree illustrates next-token **prediction**.

The facts we will use

Supplied facts

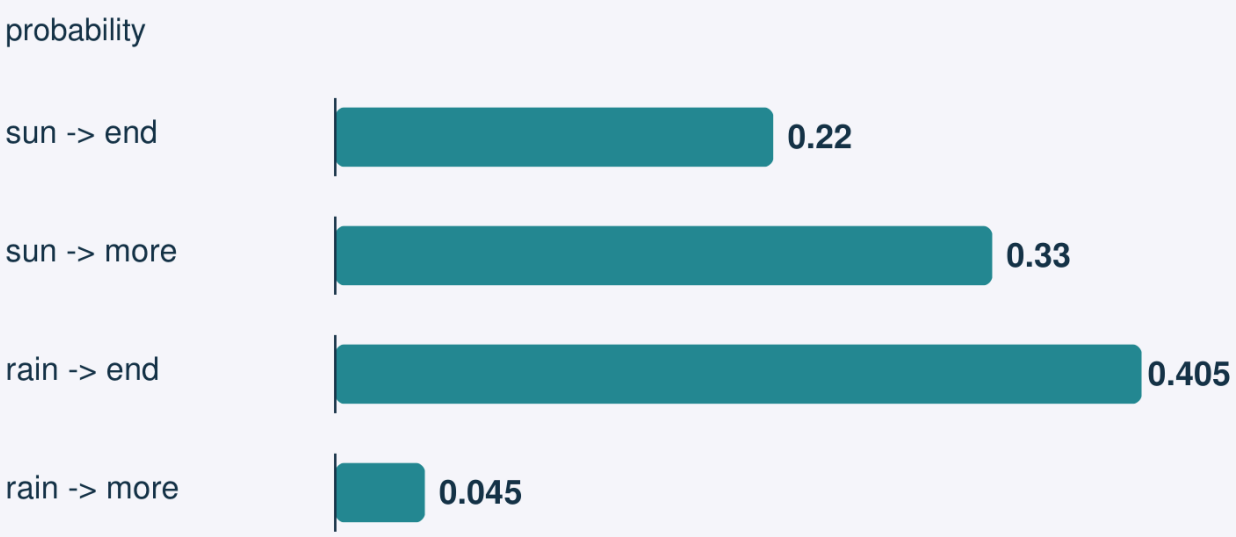
First token probabilities: sun 0.55, rain 0.45. Given sun: end 0.4, more 0.6. Given rain: end 0.9, more 0.1.

Supplied facts

These are complete toy branches; end paths stop, more paths do not.

AI ACADEMY / AI-3/2

Joint probabilities of the four toy branches



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-3/2

A Distribution of Possibilities

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. First distribution

sun 0.55, rain 0.45: greedy first choice is sun.



2. Sun branches

sun -> end:0.55 x 0.4=0.22; sun -> more:0.55 x 0.6=0.33.



3. Rain branches

rain -> end:0.45 x 0.9=0.405; rain -> more:0.45 x 0.1=0.045.



4. Compare end paths

rain -> end is the more probable of the two complete end paths.

CHANGE ONE THING

Raise end-given-sun to 0.8 and lower more-given-sun to 0.2.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. First distribution	sun 0.55, rain 0.45: greedy first choice is sun.
2. Sun branches	sun→end:0.55×0.4=0.22; sun→more:0.55×0.6=0.33.
3. Rain branches	rain→end:0.45×0.9=0.405; rain→more:0.45×0.1=0.045.
4. Compare end paths	rain→end is the more probable of the two complete end paths.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Conditional probabilities describe what follows a particular prefix. Multiplying along a path gives its joint probability.

The four branch products sum to 1, but the two end paths sum to 0.625 because more branches have not ended. Greedy decoding selects sun then more; it does not guarantee the most likely complete sequence.

What this example does not establish

Keep the stated limit

These word labels and tiny tables are illustrative; path probability is not a measure of real-world truth.

Change one thing in the pictured case

Change one thing

Change: Raise end-given-sun to 0.8 and lower more-given-sun to 0.2.

Predicted result

Prediction: sun→end becomes 0.44, beating unchanged rain→end 0.405.

Why the result follows

Reason: Changing one conditional **distribution** changes its path products while the first-token probabilities remain fixed.

Ask while showing the picture

Ask while showing the picture · 8

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate A Distribution of Possibilities, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: A toy next-token **distribution** is rain .5, sun .3, snow .2.

2. State the decision to explain

Write the question to explain: Check total and interpret the largest value.

3. Reveal the worked reasoning

Reveal after **prediction**s: Total is 1. Rain is most likely, not guaranteed under sampling. Token probability is not whole-sentence truth.

Hold back the independent task

Keep for an independent attempt: A toy next-token **distribution** assigns cat 0.5, dog 0.3 and bird 0.2. Check the total and name the most likely token. Must it be selected every time when sampling?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant

details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: red,end 0.4*0.9 = 0.36; blue,end 0.6*0.5 = 0.3. Blue is the preferred first token but red,end has the larger specified end-path probability.

Exit: the complement includes unknown, and model next-token probabilities do not measure observed room-state frequency.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and distinguish a numerical claim from its interpretation.
7 to 18 minutes	Read section 1 and complete W1; ask for a reason alongside each answer.
18 to 32 minutes	Trace the worked case in section 2 and complete W2.
32 to 45 minutes	Discuss the diagram and complete W3 in pairs; record support.
Session 2 start to 8 minutes	Complete the guided response before sharing the answer.
8 to 20 minutes	Read the remaining sections and trace another constructed case.
20 to 35 minutes	Collect W4 independently, requiring calculations and a bounded conclusion.

Stage	Teacher action
35 to 45 minutes	Discuss W5 and the exit response. Provide feedback and a fresh retry.

Teach the connected idea

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and distinguish a numerical claim from its interpretation.
7 to 18 minutes	Read section 1 and complete W1; ask for a reason alongside each answer.
18 to 32 minutes	Trace the worked case in section 2 and complete W2.
32 to 45 minutes	Discuss the diagram and complete W3 in pairs; record support.
Session 2 start to 8 minutes	Complete the guided response before sharing the answer.
8 to 20 minutes	Read the remaining sections and trace another constructed case.
20 to 35 minutes	Collect W4 independently, requiring calculations and a bounded conclusion.
35 to 45 minutes	Discuss W5 and the exit response. Provide feedback and a fresh retry.

Reduce help deliberately

Completion practice

Completion task: Compute ready,end; busy,end; unknown,end and their combined probability.

Reasoning to listen for

Teacher check: Teacher explanation: Path probabilities 0.4,0.15,0.2; combined probability of ending at step two is 0.75.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Next token **distribution**s. Proposed explanation: “Conditional products assume independence”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> #### Expected repair
>
> Expected correction: Use a different **conditional probability** for each expanded prefix.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Connected teaching cycle: Embeddings and position

Target: Look up token **embedding**s and add explicit **position vector**s

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: Compute representations for red blue and blue red under the stated position table.

> #### Reasoning to listen for
>
> Teacher check: Teacher explanation: red blue rows [1,0],[2,1]; blue red rows [0,1],[3,0]. Addition is component by component.

Individual check: For **Embedding**s and position, use its worked case to evaluate this proposed explanation: “A **position vector** guarantees every summary preserves order”. Explain the deciding step.

Expected repair

Expected correction: The supplied sum counterexample loses the difference.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

Pictorial practice question

Why is 0.625 not a broken probability sum?

Reasoned teacher answer

It counts only the two paths ending now; the remaining 0.375 belongs to sun→more and rain→more.

Changed-case reasoning: sun→end becomes 0.44, beating unchanged rain→end 0.405. Changing one conditional **distribution** changes its path products while the first-token probabilities remain fixed.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the A Distribution of Possibilities investigation and use the first unsupported step to

> choose a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Build a weighted token spinner, **sample** twenty times, tally outcomes, and compare observed counts with expected relative weights.

> What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Conditional products assume independence: Use a different **conditional probability** for each expanded prefix.

Greedy decoding maximises every full sequence: The supplied two-step counterexample disproves that general claim.

Not ready means busy: The stated alternatives also include unknown.

Check these misconceptions

A **position vector** guarantees every summary preserves order: The supplied sum counterexample loses the difference.

IDs are **embedding** coordinates: IDs select rows; vector components are separate data.

Any vector widths can be added: **Elementwise addition** requires equal widths.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

TEACHER GUIDE · 5 OF 9

Assess and explain the answers

> Use the A Distribution of Possibilities questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Total is 1. Rain is most likely, not guaranteed under sampling. Token probability is not whole-sentence truth.

Independent question

Independent question · withhold the key

A toy next-token **distribution** assigns cat 0.5, dog 0.3 and bird 0.2. Check the total and name the most likely token. Must it be selected every time when sampling?

Expected independent answer

Teacher key · Expected independent answer

The probabilities sum to 1.0. Cat is most likely, but sampling can also select dog or bird.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “The largest probability makes every other outcome impossible.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

> #### Teacher key · Original transfer answer
>
> No. Five is the expected count under independent sampling from this fixed **distribution**, not a required result. Token probability describes **continuation** choices, not the truth of a weather claim.
>
> Actual forecasting requires relevant **observation**s and evaluation.

Explanations for the additional workbook examples

W1 Validate the table

Workbook question · W1

Check 0.5,0.3,0.2 as probabilities. Interpret their sum and the complement of ready.

Teacher reasoning · W1

Teacher explanation: Entries are nonnegative and sum to 1. Not ready has mass 0.3+0.2 = 0.5, including busy and unknown.

W2 Calculate the paths

Workbook question · W2

Compute ready,end; busy,end; unknown,end and their combined probability.

Teacher reasoning · W2

Teacher explanation: Path probabilities 0.4,0.15,0.2; combined probability of ending at step two is 0.75.

W3 Test a greedy claim

Workbook question · W3

Use A=0.6,B=0.4 and end conditional probabilities 0.51 and 1. Compare greedy A,end with B,end.

Teacher reasoning · W3

Teacher explanation: A,end probability 0.306 versus B,end 0.4. Greedy next-token choice does not guarantee the largest complete-path probability.

W4 Analyse a new table independently

Workbook question · W4

First tokens go=0.4, wait=0.35, ask=0.25. End probabilities given them are 0.5,0.8,1 respectively; each branch has a complementary more token. Compute all three end-path probabilities and their sum.

Which first token is greedy, which end path is largest, and does either verify a real status?

Teacher reasoning · W4

Teacher explanation: End-path probabilities 0.2,0.28,0.25; sum 0.73. Greedy first token is go, but wait,end is the largest of the specified end paths. Neither establishes a real status without appropriate **evidence**.

W5 Normalise with a stated rule

Workbook question · W5

Weights are ready 6,busy 3,unknown 1. Convert them into probabilities. What if a weight is -1?

Teacher reasoning · W5

Teacher explanation: Total 10 yields 0.6,0.3,0.1. Negative weights violate this nonnegative-weight normalisation contract; reject rather than quietly normalise them.

Embeddings and position — W1 Look up vectors

For red blue name the IDs, **embedding** rows and shape before adding positions.

> **Teacher reasoning**
>
> Teacher explanation: IDs [1,2]; rows [1,0],[0,1]; shape two positions by two components.

Embeddings and position — W2 Add positions

Compute representations for red blue and blue red under the stated position table.

> **Teacher reasoning**
>
> Teacher explanation: red blue rows [1,0],[2,1]; blue red rows [0,1],[3,0]. Addition is component by component.

Embeddings and position — W3 Audit the sum

> Sum the rows for both sequences. Does the sum establish that the original text had the same order?

> **Teacher reasoning**
>
> Teacher explanation: Both sums [3,1]. Summing discards the ordered-row difference; equal summaries do not prove equal source sequences.

Embeddings and position — W4 Trace new vectors independently

Embeddings A=[1,2], B=[3,0], position 0=[0,1], position 1=[1,0]. Compute represented rows and sums for A B and B A. Explain what the equal sums mean.

Teacher reasoning

Teacher explanation: A B rows [1,3],[4,0]; B A rows [3,1],[2,2]. Both sums [5,3]; the summaries lose the ordering difference while the ordered row representations differ.

Embeddings and position — W5 Handle a shape failure

An **embedding** has width two but its **position vector** has width three. The **input** also contains a third position absent from the table. Specify honest handling.

Teacher reasoning

Teacher explanation: Reject mismatched vector widths and unsupported position explicitly. Do not truncate, wrap or invent values unless a separate documented contract defines that transformation.

Assess independent transfer

Independent question · withhold the key

Selected case: First tokens go=0.4, wait=0.35, ask=0.25. End probabilities given them are 0.5,0.8,1 respectively; each branch has a complementary more token. Compute all three end-path probabilities and their sum.

Which first token is greedy, which end path is largest, and does either verify a real status?

Selected case answer

Teacher key · Selected case answer

Teacher explanation: End-path probabilities 0.2,0.28,0.25; sum 0.73. Greedy first token is go, but wait,end is the largest of the specified end paths. Neither establishes a real status without appropriate **evidence**.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect A Distribution of Possibilities to Verified Study Buddy, adapt support and plan a later changed-case check.

Verified Study Buddy

Explain that Study Buddy responses come from possibilities, so important claims need checks.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Provide one short approved passage; separate instruction, **input**, **output** and **evidence**. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 3: Tokens and Context. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Provide labelled intermediate columns and allow a calculator. Ask students to state the meaning and denominator of a value before interpreting it.

Extension: implement the classroom calculation locally and test boundary cases without using personal data.

W4 requires 0.2,0.28,0.25, sum 0.73, greedy go and largest end path wait,end with an **evidence** limit. Fresh retry: first A=0.45,B=0.3,C=0.25; end conditionals 0.4,0.9,1.

Products 0.18,0.27,0.25 sum 0.70; greedy A, largest end path B,end.

Week 4 distinguishes greedy choice from sampling and computes the effect of temperature on a fixed toy **distribution**.

Readiness and reassessment for Embeddings and position

Provide labelled intermediate columns and allow a calculator. Ask students to state the meaning and denominator of a value before interpreting it.

Extension: implement the classroom calculation locally and test boundary cases without using personal data.

W4 requires both row pairs and shared sum [5,3], with an information-loss explanation. Fresh retry: A=[2,0], B=[0,4], positions [1,0] and [0,2]. A B gives [3,0],[0,6]; B A gives [1,4],[2,2]; sums [3,6].

Plan the delayed return

Suggested return points in this level: ai-3/3, ai-3/5, ai-3/8. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Explain Study Buddy generation as conditional choices, while checking the finished factual claims against sources.

Prepare, teach and assess

> Use this preparation summary alongside the detailed A Distribution of Possibilities lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Check whether a next-token table is a valid **probability distribution**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Compute ready,end; busy,end; unknown,end and their combined probability.

Supported practice answer

Reasoned practice answer: Teacher explanation: Path probabilities 0.4,0.15,0.2; combined probability of ending at step two is 0.75.

Check every learner before discussion: Use the worked case for Next token **distribution**s. Proposed explanation: “Conditional products assume independence”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Use a different **conditional probability** for each expanded prefix.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Raise end-given-sun to 0.8 and lower more-given-sun to 0.2. Require a **prediction** and a reason before comparing. Expected result: sun→end becomes 0.44, beating unchanged rain→end 0.405.

Reason: Changing one conditional **distribution** changes its path products while the first-token probabilities remain fixed.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

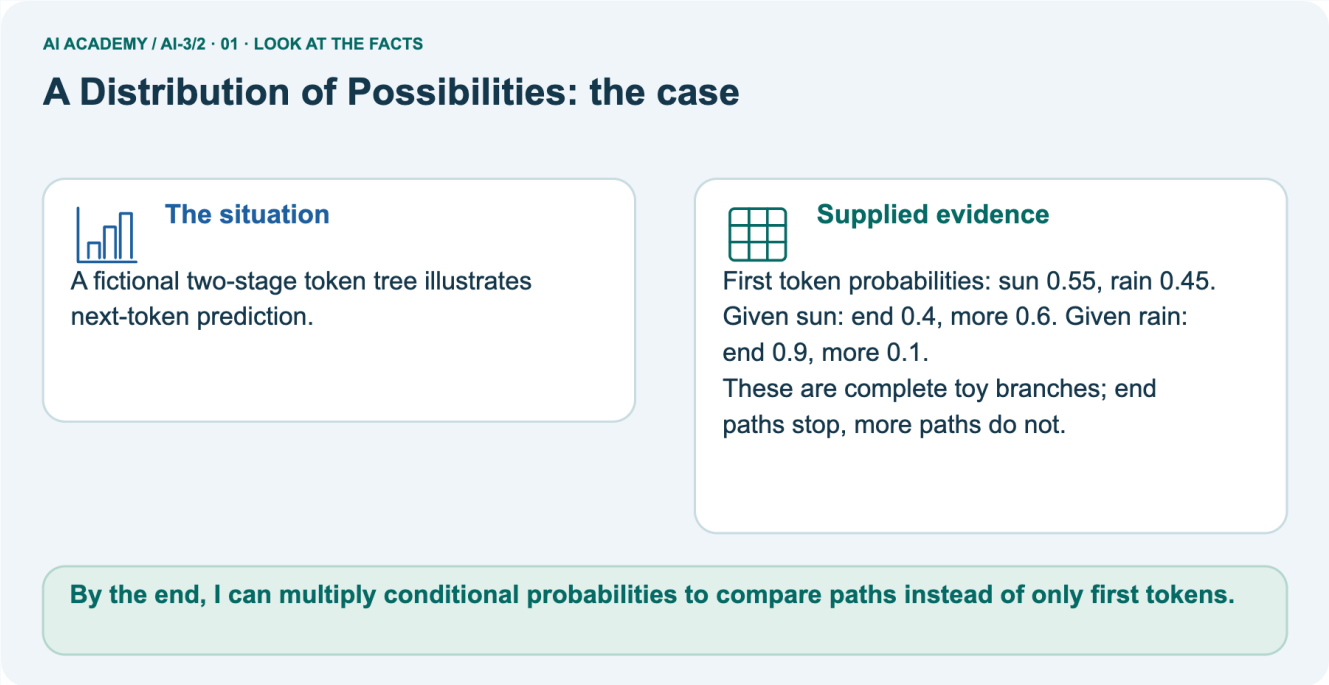
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one A Distribution of Possibilities graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. A Distribution of Possibilities: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional two-stage token tree illustrates next-token **prediction**. Supplied facts: First token probabilities: sun 0.55, rain 0.45. Given sun: end 0.4, more 0.6. Given rain: end 0.9, more 0.1.

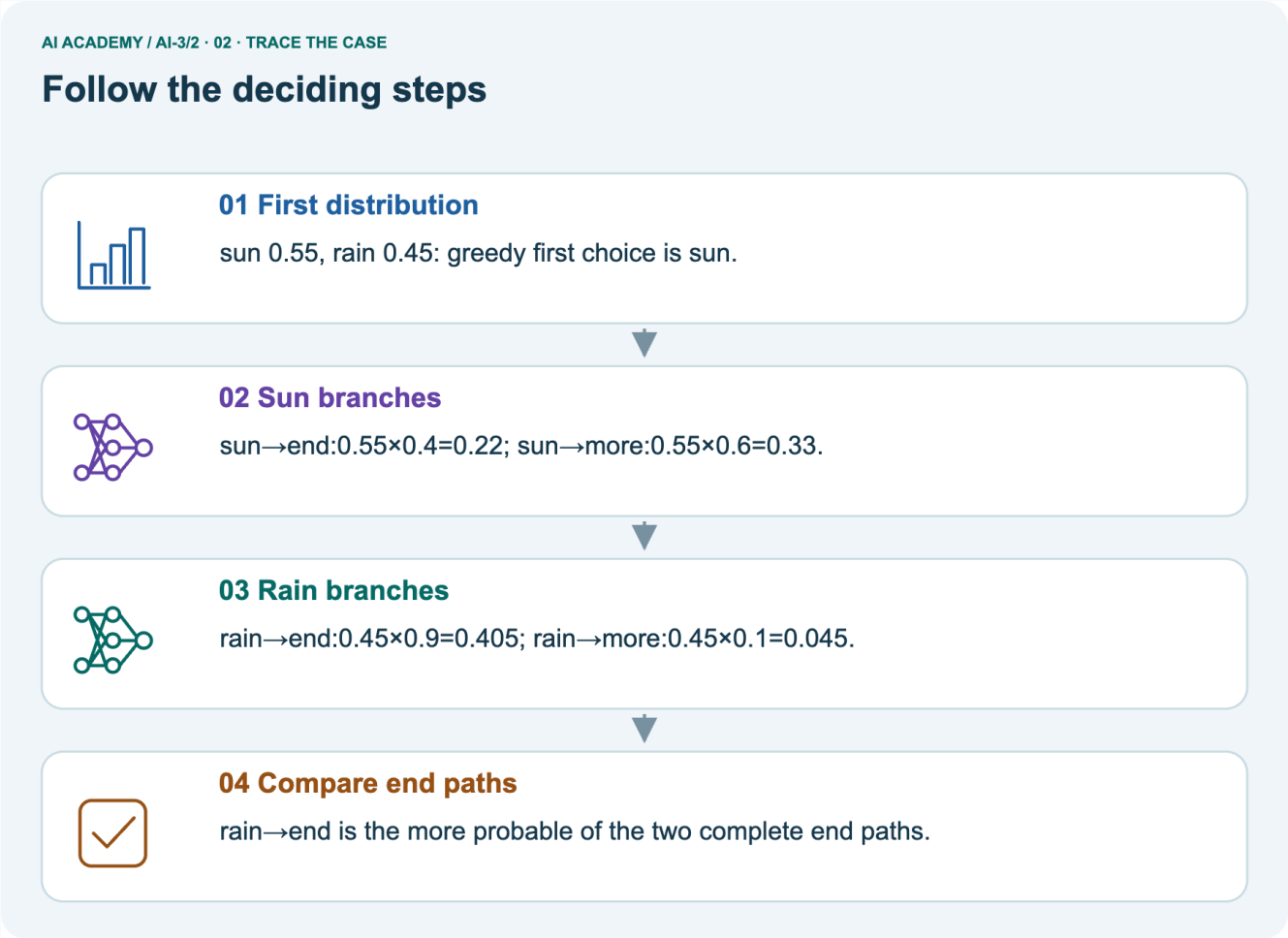
These are complete toy branches; end paths stop, more paths do not.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

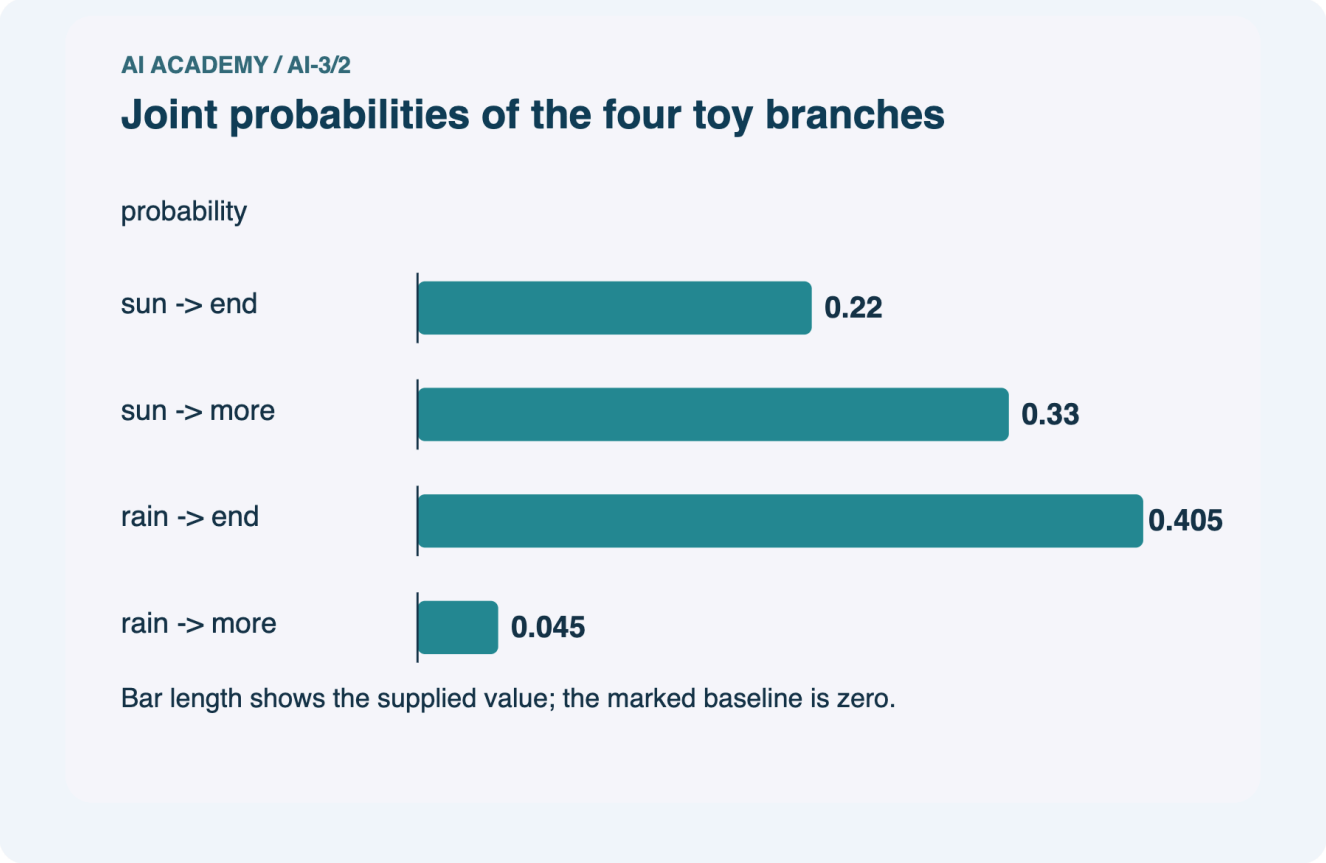
First **distribution**: sun 0.55, rain 0.45: greedy first choice is sun. → Sun branches: sun→end:0.55×0.4=0.22; sun→more:0.55×0.6=0.33. → Rain branches: rain→end:0.45×0.9=0.405; rain→more:0.45×0.1=0.045. → Compare end paths: rain→end is the more probable of the two complete end paths.

> Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Joint probabilities of the four toy branches



Follow the labelled picture with the matching explanation. Open the vector diagram

Joint probabilities of the four toy branches. Units: probability. sun→end: 0.22. sun→more: 0.33. rain→end: 0.405. rain→more: 0.045.

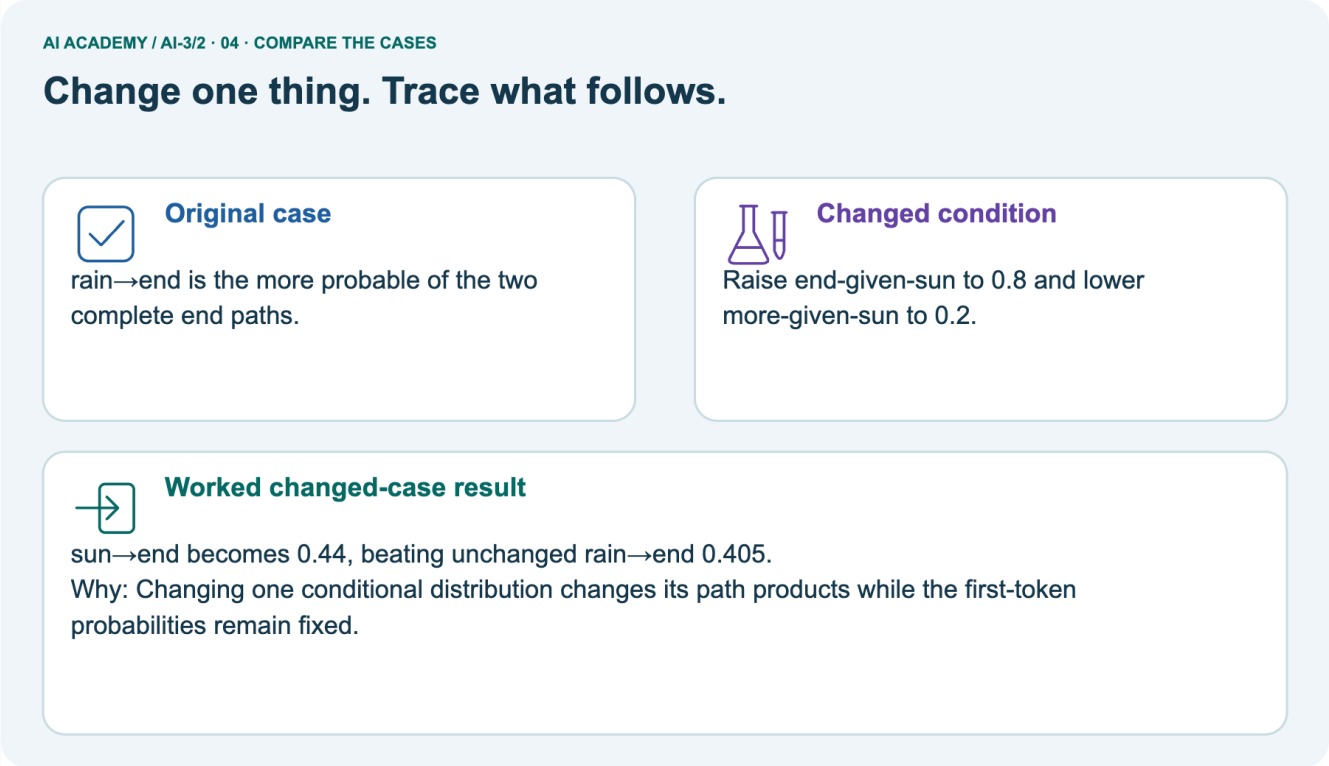
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: rain→end is the more probable of the two complete end paths. Changed condition: Raise end-given-sun to 0.8 and lower more-given-sun to 0.2.

Expected result: sun→end becomes 0.44, beating unchanged rain→end 0.405. Reason: Changing one conditional **distribution** changes its path products while the first-token probabilities remain fixed.

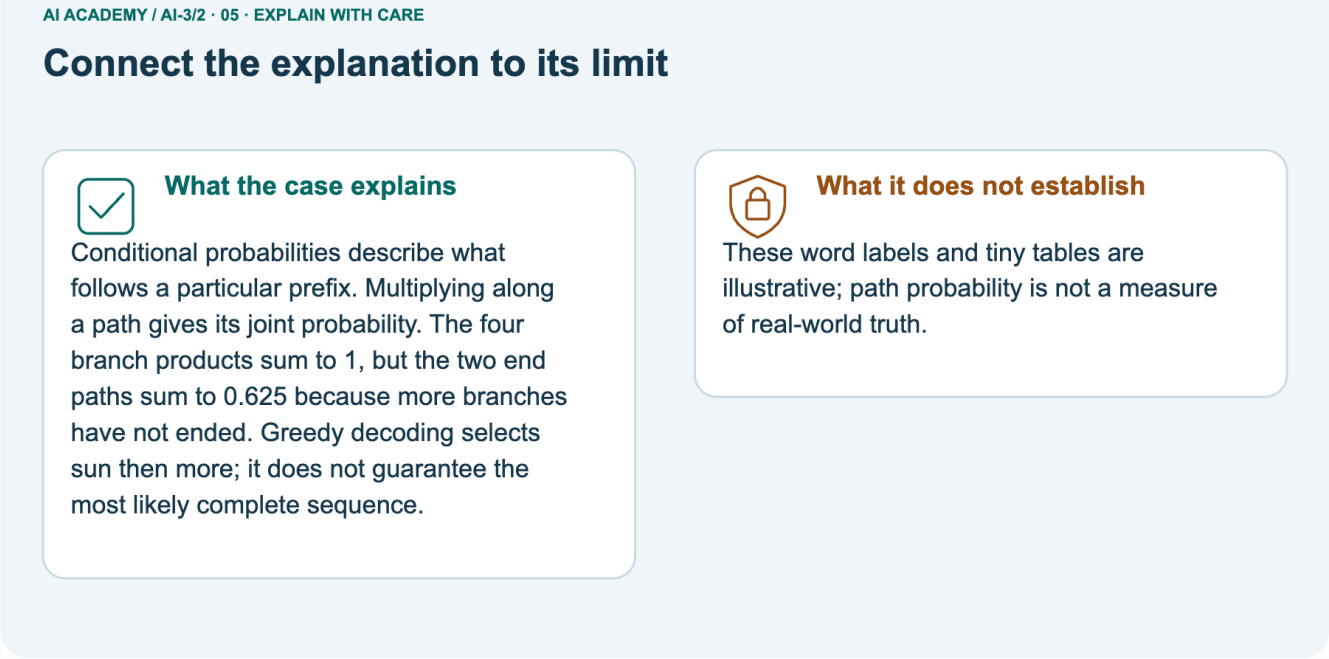
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Changing one conditional **distribution** changes its path products while the first-token probabilities remain fixed.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Conditional probabilities describe what follows a particular prefix. Multiplying along a path gives its joint probability.

The four branch products sum to 1, but the two end paths sum to 0.625 because more branches have not ended. Greedy decoding selects sun then more; it does not guarantee the most likely complete sequence.

Limit: These word labels and tiny tables are illustrative; path probability is not a measure of real-world truth.

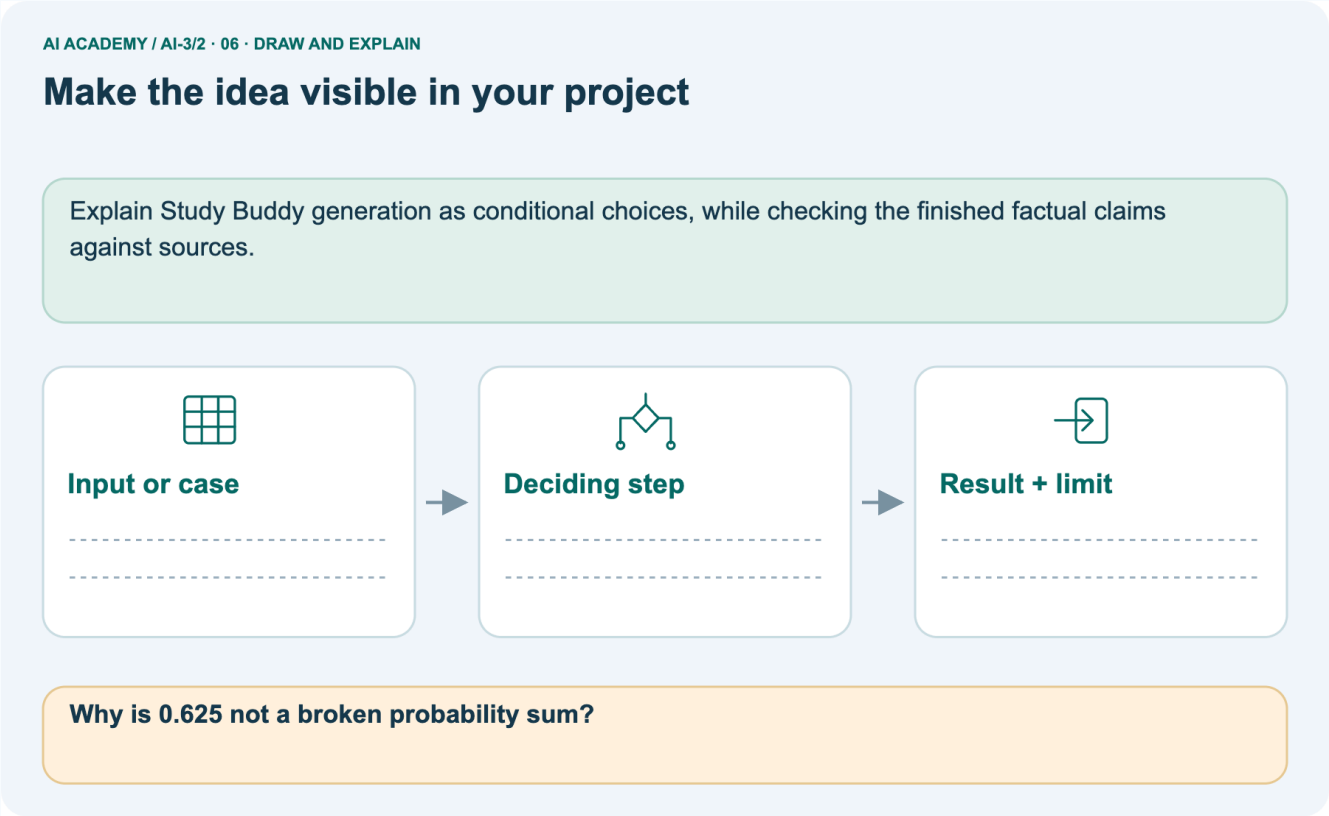
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: These word labels and tiny tables are illustrative; path probability is not a measure of real-world truth.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain Study Buddy generation as conditional choices, while checking the finished factual claims against sources. Draw your **input**, deciding step, result and limit.

Supported question: Why is 0.625 not a broken probability sum?

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: It counts only the two paths ending now; the remaining 0.375 belongs to sun→more and rain→more.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.